StreamWatch: Day 1 Dimensional Design Blueprint

1. Fact Table Grain Declaration

- **Declared Grain:** One row per individual watch session (event_id).
- **Definition:** Every single row in our main fact table represents one time a user sat down to stream a specific video title, on a specific date, using a specific type of device.

2. Column Classification Table

Source Column Name	Classification / Bucket	Destination Table	Plain English Rationale
event_id	Degenerate Dimension	fct_watch_events	The unique ID for the session. We leave it directly in the fact table so we can track individual rows without wasting time creating an extra empty table for it.
user_id	Foreign Key	fct_watch_events	The connecting ID link used to join this session to the user information table.
user_name	Dimension Attribute	dim_users	Text description showing the name of the user.
user_country	Dimension Attribute	dim_users	Text description showing where the user lives.
user_subscription_tier	Dimension Attribute	dim_users	Text description of the account type (Free, Basic, Premium).
content_id	Foreign Key	fct_watch_events	The connecting ID link used to join this session to the video information table.
content_title	Dimension Attribute	dim_content	The actual name or title of the video.
content_genre	Dimension Attribute	dim_content	The category or style of the video (e.g., Comedy, Sci-Fi).
content_release_date	Dimension Attribute	dim_content	The specific calendar date when the video originally came out.
content_duration_minutes	Dimension Attribute	dim_content	The full runtime of the video in minutes. It describes the content itself, so it goes in the content table.
watch_date	Foreign Key / Date	fct_watch_events	The date the streaming session actually happened. Links to our time tracking.
device_type	Degenerate Dimension	fct_watch_events	A simple text description of the hardware used (Mobile, TV, etc.) left directly in the fact table for easy filtering.
minutes_watched	Additive Measure	fct_watch_events	A clean number showing how long the session lasted. We can easily add these numbers up or find the average.
completed_flag	Stored Measure	fct_watch_events	A pre-calculated 1 or 0 flag showing if the user finished the video. We store it as a number so analysts can easily sum it.
rating_given	Nullable Measure	fct_watch_events	The score (1 to 5) a user left. It is left blank (NULL) if the user decided not to rate the video.
days_since_release	Derived Measure	fct_watch_events	A calculated number showing how many days passed between the video's release date and the actual watch date.

3. Position & Justification on the Four Design Tensions

Tension 1: device_type (Should it get its own table or stay in the main table?)

- **Position:** Keep it directly in the main fact table (fct_watch_events).

• **Justification:** There are only four or five device options in the entire dataset (like Mobile, TV, Desktop, and Tablet). Because there is no extra detailed information to track for these devices (like the operating system version or brand names), creating a brand new dimension table is just extra work. Keeping it as text directly in the fact table gives analysts the exact same insights without forcing them to write an extra table join.

Tension 2: completed_flag vs. minutes_watched (Should we store the flag or calculate it with math later?)

• **Position:** Keep the completed_flag stored exactly as it is in the fact table.

• **Justification:** If we try to drop this column and calculate completions using a strict math formula (like checking if minutes watched equals 100% of the video duration), our reports will break. The data shows that a session is marked as "completed" around the 85–90% mark because people naturally turn off a video when the credits start rolling. If we rely strictly on a math formula, we will mistakenly mark 978 finished sessions as incomplete. Storing the flag as-is keeps our numbers accurate and saves analysts from writing messy calculations.

Tension 3: What to do with rating_given (One combined table vs. two separate tables?)

• **Position:** Keep it as a column that allows blank spaces (NULL) inside the main fact table.

• **Justification:** In this specific dataset, users leave a rating right at the end of their watch session. Splitting ratings into a completely separate table means we have to build, test, and maintain a whole new database pipeline just to hold one single column. While it means our main table will have empty spaces when users choose not to leave a review, it keeps our project structure simple, clean, and much faster to build.

Tension 4: content_release_date (Leave it as a plain detail or calculate the time gap?)

• **Position:** Keep it as a plain detail in the content table, but calculate days_since_release directly inside our fact table layer.

• **Justification:** The date a movie came out belongs inside the content information table because that date never changes. However, calculating the difference between the watch date and the release date gives business analysts a massive advantage. It allows them to instantly run reports to see if a new video gets popular the day it drops, or if it takes months to gain traction.